

# POLI 784 - Problem Set #3

Kieran Fitzmason, Jacob Seigel, Xiangyong Zhang

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## Question 3

In this exercise, we analyze data from a study by [Santoro and Broockman \(2022\)](#) examining the effect of conversations between strangers of opposing political parties on measures of polarization. During their conversation, subjects talked about what their perfect day would look like. Subjects assigned to treatment ( $D_i = 1$ ) were informed prior to the start of the conversation that their partner was an out-partisan, while those assigned to control ( $D_i = 0$ ) were not informed of this. The primary outcome of interest  $Y_i$  was self-reported warmth towards outpartisans measured on a feeling thermometer.

```
# Load required packages
library("tidyverse")
library("estimatr")
library("sandwich")
library("interflex")
library("car")
library("glue")

# Set random seed for reproducibility
set.seed(42)
```

### Part A

Here, we compute an ATE point estimate using the Neyman estimator and corresponding analytical Neyman variance estimate. The formulae for these quantities is as follows:

$$\hat{\tau}_{DIM} = \bar{Y}_1 - \bar{Y}_0 = \frac{1}{n_1} \sum_{i:D_i=1} Y_i - \frac{1}{n_0} \sum_{i:D_i=0} Y_i \quad (1)$$

$$\hat{V}[\hat{\tau}_{DIM}] = \frac{s_1^2}{n_1} + \frac{s_0^2}{n_0} \quad (2)$$

where  $n_1$  and  $n_0$  are the number of units assigned to treatment and control, and  $s_1^2$  and  $s_0^2$  are the sample variances of the outcome variable in the treatment and control groups.

We subsequently compare these estimates to those produced using OLS regression with no covariates and the HC2 variance estimator.

```
# Load data
df <- read.csv("data-2.csv")

# Rename outcome and treatment columns
colnames(df)[1:2] <- c("Y","D")

# Calculate ATE point estimate using Neyman estimator
n1 = sum(df$D)
n0 = sum(1-df$D)
tau_DIM <- mean(df$Y[df$D==1])-mean(df$Y[df$D==0])
var_DIM <- var(df$Y[df$D==1])/n1 + var(df$Y[df$D==0])/n0

# Calculate ATE using OLS and HC2 variance estimator
model <- lm_robust(Y ~ D, data = df, se_type = "HC2")
tau_OLS <- model$coefficients["D"]
var_HC2 <- model$std.error["D"]^2

# Display to console
glue("Neyman ATE point estimate: {round(tau_DIM,6)}")
```

Neyman ATE point estimate: 0.213034

```
glue("Neyman variance: {round(var_DIM,6)}")
```

Neyman variance: 0.008334

```
glue("OLS ATE point estimate: {round(tau_OLS,6)}")
```

OLS ATE point estimate: 0.213034

```
glue("HC2 variance: {round(var_HC2,6)}")
```

HC2 variance: 0.008334

As shown above, we obtained identical estimates of the ATE and corresponding variance when using the Neyman estimator and OLS/HC2 estimator. This is not surprising, as the two approaches are algebraically equivalent in the case of a single covariate representing a binary treatment status. A proof of this is provided by [Samii and Aronow \(2012\)](#), which we briefly summarize below.

Assume we are fitting the following model:

$$Y_i = \alpha + \beta D_i + e_i \quad (3)$$

When  $D_i$  is a binary variable, the OLS estimates of  $\alpha$  and  $\beta$  reduce to:

$$\hat{\alpha} = \bar{Y}_0 \quad (4)$$

$$\hat{\beta} = \bar{Y}_1 - \bar{Y}_0 \quad (5)$$

It can also be shown that for a binary treatment, the HC2 variance estimator reduces to:

$$\hat{V}_{HC2}[\hat{\beta}] = \frac{1}{n_1} \frac{1}{n_1 - 1} \sum_{i:D_i=1} \hat{e}_i^2 + \frac{1}{n_0} \frac{1}{n_0 - 1} \sum_{i:D_i=0} \hat{e}_i^2 \quad (6)$$

where  $\hat{e}_i$  represents the regression residual. This residual can be rewritten as:

$$\hat{e}_i = Y_i - \hat{\alpha} - \hat{\beta} D_i = Y_i - \bar{Y}_0 - (\bar{Y}_1 - \bar{Y}_0) D_i \quad (7)$$

Thus,  $\hat{e}_i = Y_i - \bar{Y}_1$  when  $D_i = 1$  and  $\hat{e}_i = Y_i - \bar{Y}_0$  when  $D_i = 0$ . By substituting these expressions for  $\hat{e}_i$  into Equation 6, we can rewrite the HC2 variance as:

$$\hat{V}_{HC2}[\hat{\beta}] = \frac{1}{n_1} \frac{1}{n_1 - 1} \sum_{i:D_i=1} (Y_i - \bar{Y}_1)^2 + \frac{1}{n_0} \frac{1}{n_0 - 1} \sum_{i:D_i=0} (Y_i - \bar{Y}_0)^2 = \frac{s_1^2}{n_1} + \frac{s_0^2}{n_0} \quad (8)$$

Which is equivalent to the formula for the Neyman variance estimate shown in Equation 2.

## Part B

Often we are fitting a model of the following form:

$$y_i = \mu + \tau d_i + \mathbf{x}_i^\top + e_i$$

The Frisch-Waugh-Lovell (FWL) theorem states that we can estimate  $\tau$  via the following procedure:

1. Regress  $y_i$  on  $\mathbf{x}_i$  and save the residuals  $\hat{e}_{Y_i}$ .
2. Regress  $d_i$  on  $\mathbf{x}_i$  and save the residuals  $\hat{e}_{D_i}$ .
3. Regress  $\hat{e}_{Y_i}$  on  $\hat{e}_{D_i}$ . The coefficient of  $\hat{e}_{D_i}$  will be  $\hat{\tau}$ .

Here, we show that the estimate of  $\hat{\tau}$  produced via the FWL procedure is equivalent to the estimate produced via multiple linear regression (MLR).

```
## Clean columns corresponding to covariates

# Began convo: Drop this variable since value is the same for everyone.
df <- df[, !(names(df) == "began_convo")]

# Gender: Set female to zero, everyone else as one.
df$gender <- 1 - as.numeric(df$gender == "Female")

# Note: we assume that all other covariates
# are numeric and continuous, and thus do not modify them.

## Estimate tau via MLR
MLR_model <- lm_robust(Y ~ D +
                      age +
                      gender +
                      partyideology +
                      pre_therm_voter_outparty_rescaled,
                      data = df,
                      se_type="HC2")

tau_MLR <- MLR_model$coefficients["D"]

## Estimate tau via FWL procedure

# Step 1: Regress Y on X
```

```

step1_model <- lm(Y ~ age +
                  gender +
                  partyideology +
                  pre_therm_voter_outparty_rescaled,
                  data = df)

df$FWL_resid_Y <- step1_model$residuals

# Step 2: Regress D on X
step2_model <- lm(D ~ age +
                  gender +
                  partyideology +
                  pre_therm_voter_outparty_rescaled,
                  data = df)

df$FWL_resid_D <- step2_model$residuals

# Step 3: Regress residuals from step 1 on residuals from step 2
FWL_model <- lm_robust(FWL_resid_Y ~ FWL_resid_D, data = df, se_type="HC2")
tau_FWL <- FWL_model$coefficients["FWL_resid_D"]

# Display to console
glue("Multiple linear regression: tau = {round(tau_MLR,6)}")

```

Multiple linear regression: tau = 0.326836

```
glue("Frisch-Waugh-Lovell procedure: tau = {round(tau_FWL,6)}")
```

Frisch-Waugh-Lovell procedure: tau = 0.326836

## Part C

Here, we use the residuals computed as part of the FWL procedure to examine the effect of  $D_i$  on  $Y_i$  before and after adjusting for covariates.

```

# Plot effect of D on Y after adjusting for X
p1 <- ggplot(df, aes(x = FWL_resid_D, y = FWL_resid_Y)) +
  geom_point() +
  geom_smooth(method = "lm_robust", method.args=list(se_type="HC2")) +
  xlim(-0.7,0.7) +

```

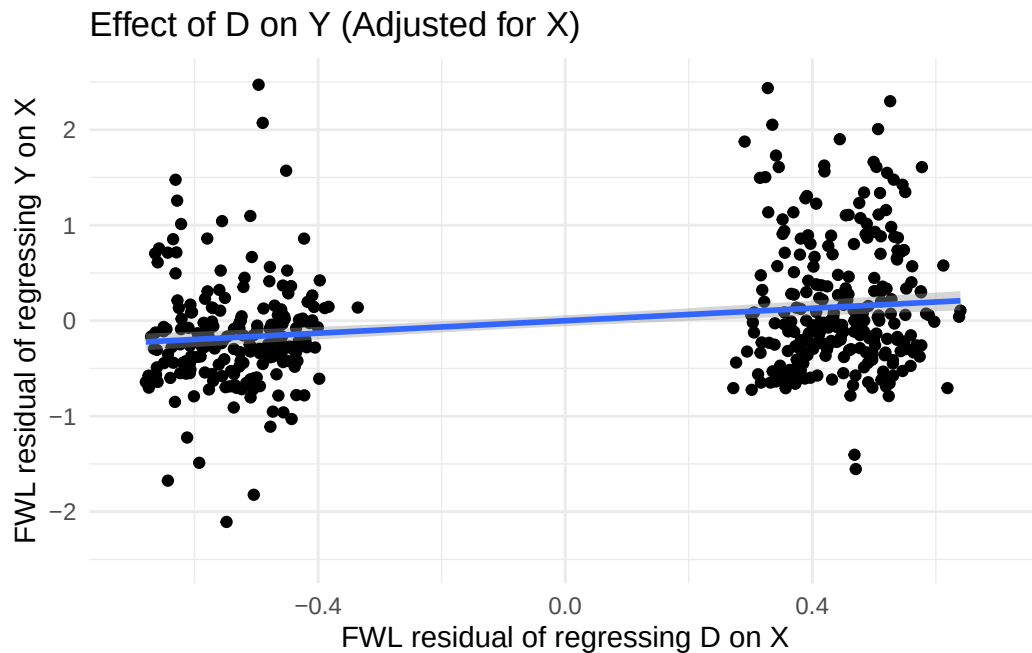
```

ylim(-2.5,2.5) +
labs(
  x = "FWL residual of regressing D on X",
  y = "FWL residual of regressing Y on X",
  title = "Effect of D on Y (Adjusted for X)"
) +
theme_minimal()

# Display to console
p1

```

`geom\_smooth()` using formula = 'y ~ x'



```

# Plot effect of D on Y without adjusting for X
p2 <- ggplot(df, aes(x = D, y = Y)) +
  geom_point() +
  geom_smooth(method = "lm_robust", method.args=list(se_type="HC2")) +
  xlim(-0.2,1.2) +
  ylim(-2.5,2.5) +
  labs(
    x = "Treatment (D)",

```

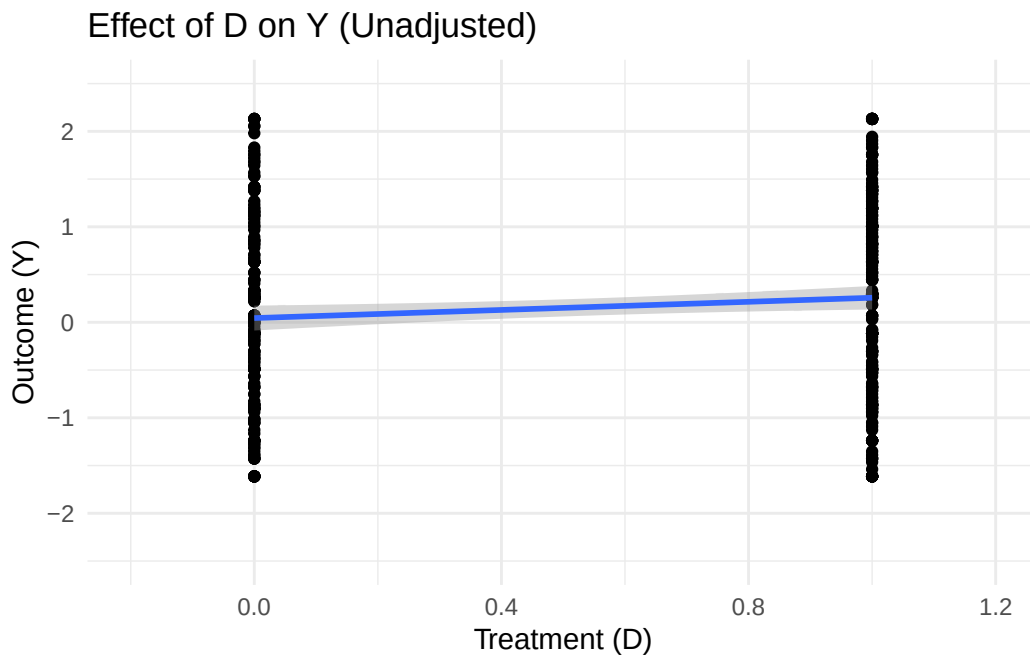
```

  y = "Outcome (Y)",
  title = "Effect of D on Y (Unadjusted)"
) +
  theme_minimal()

# Display to console
p2

```

``geom_smooth()`` using formula = 'y ~ x'



One notable difference in these two scatterplots is that the standard error on the regression line appears to be smaller for the plot that adjusts for covariates via the FWL procedure. This suggests that even in randomized experiments, it can often be helpful to adjust for baseline covariates in order to more efficiently estimate effects.

## Part D

Here, we fit a Lin regression model that captures interactions between the treatment and covariates. The estimated treatment effect and associated 95% confidence interval is subsequently compared against those obtained via multivariate OLS and bivariate OLS.

```

# Fit a Lin regression model
lin_model <- lm_lin(Y ~ D,
                   covariates = ~age +
                     gender +
                     partyideology +
                     pre_therm_voter_outparty_rescaled,
                   data = df,
                   se_type = "HC2")

lin_estimate <- lin_model$coefficients["D"]
lin_LB <- lin_model$conf.low["D"]
lin_UB <- lin_model$conf.high["D"]

# Multiple OLS
multi_OLS_estimate <- MLR_model$coefficients["D"]
multi_OLS_LB <- MLR_model$conf.low["D"]
multi_OLS_UB <- MLR_model$conf.high["D"]

# Bivariate OLS
biv_OLS_estimate <- model$coefficients["D"]
biv_OLS_LB <- model$conf.low["D"]
biv_OLS_UB <- model$conf.high["D"]

# Dump results into a dataframe that we'll use for plotting
results_df <- tibble(
  method=c("Lin","Multiple OLS","Bivariate OLS"),
  estimate=c(lin_estimate,multi_OLS_estimate,biv_OLS_estimate),
  conf.low=c(lin_LB,multi_OLS_LB,biv_OLS_LB),
  conf.high=c(lin_UB,multi_OLS_UB,biv_OLS_UB)
)

# Create coefficient plot
p <- ggplot(results_df, aes(x = estimate, y = method)) +
  geom_point(size = 3) +
  geom_errorbar(
    aes(xmin = conf.low, xmax = conf.high),
    width = 0.25
  ) +
  geom_vline(xintercept = 0, linetype = "dashed", color = "gray50") +
  labs(
    x = "Average Treatment Effect",
    y = NULL,

```

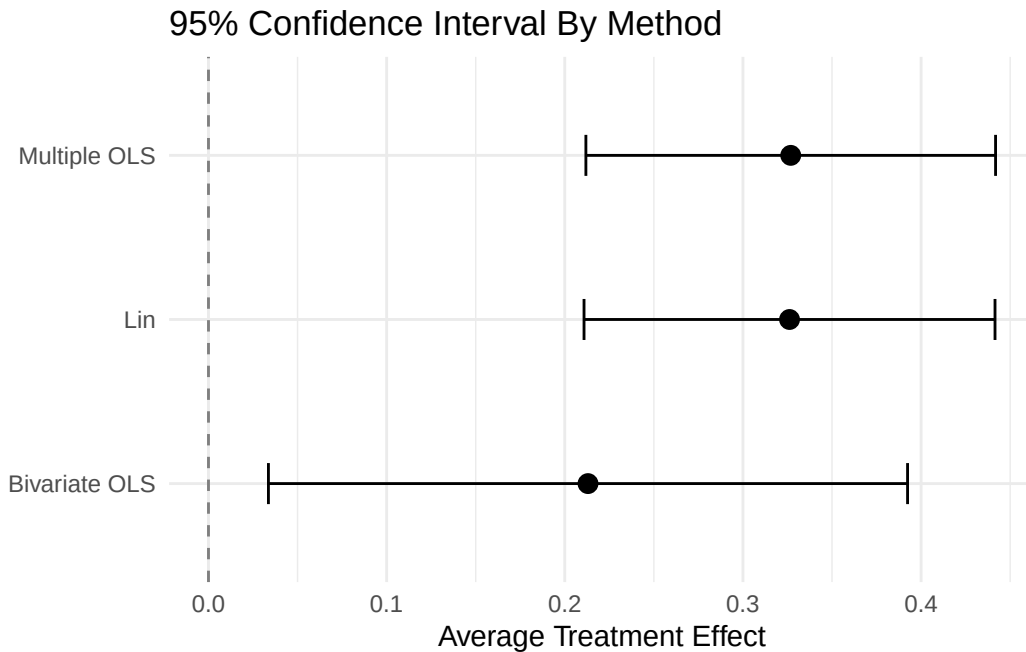


```

    title = "95% Confidence Interval By Method"
  ) +
  theme_minimal()

# Display to console
p

```



Below, we compare the three regression approaches in terms of their efficiency, unbiasedness, and plausibility of underlying assumptions. Many of our interpretations are based on the papers by [Freedman \(2008\)](#) and [Lin \(2013\)](#). Please note that we are assuming a binary treatment throughout.

### Efficiency

**Lin's Regression:** More efficient than bivariate OLS. Similar in efficiency to multivariate OLS in the above example, but in general should be more efficient than multivariate OLS because it can deal with heterogeneous treatment effects.

**Multivariate OLS:** More efficient than bivariate OLS and similar in efficiency to Lin's regression in the above example. However, it could be less efficient than one or both of these methods in other scenarios, such as in the presence of heterogeneous treatment effects or when there are nonlinear relationships between the outcome variable and pre-treatment covariates.

**Bivariate OLS:** Less efficient than Lin’s regression. In the above example, it’s also less efficient than multivariate OLS, but this may not hold in the presence of heterogeneous treatment effects.

## Unbiasedness

All three estimators require SUTVA, positivity, and strong ignorability to hold in order to produce unbiased estimates of the ATE. Additional estimator-specific requirements are listed below.

**Lin’s Regression:** Assumes a linear relationship between pre-treatment covariates and outcome, but is relatively robust to deviations from this assumption. Can accommodate heterogeneous treatment effects.

**Multivariate OLS:** Assumes a linear relationship between pre-treatment covariates and outcome. Also assumes a constant treatment effect.

**Bivariate OLS:** None. Equivalent to the Neyman difference-in-means estimator for binary treatment, so should be robust to heterogeneous treatment effects and nonlinear relationships between covariates and outcome.

## Plausibility

In our specific case, the assumption of a constant treatment effect seems implausible. For example, we suspect that participants who lie in the middle of the political spectrum may respond differently to treatment than those at the extremes, who may be less tolerant of opposing viewpoints. As such, we would recommend using Lin’s regression over multivariate OLS since the former can support heterogeneous treatment effects.

It also feels somewhat implausible to assume a linear relationship between the outcome variable (warmth towards outpartisans) and certain pre-treatment covariates such as age and party ideology. An easy way to address this issue would be to encode these variables as categorical predictors (e.g., “age 20-29”, “age 30-39”, etc.) rather than continuous predictors.

## Part E

Here, we test the joint null hypothesis that the effect of  $D_i$  on  $Y_i$  equals zero for both women and men. In Part D, we fit a Lin’s regression model that has the following form:

$$Y_i = \beta_0 + \beta_1 D_i + \dots + \beta_7 (D_i \times (G_i - \bar{G})) + \dots$$

Where  $G_i$  is a binary variable that takes on a value of zero for women and one for men and  $\bar{G}$  is the proportion of men in our sample. In this model, we can calculate conditional average treatment effects (CATE) among women and men as follows:

$$\text{CATE}_{\text{women}} = \beta_1 - \bar{G}\beta_7$$

$$\text{CATE}_{\text{men}} = \beta_1 + (1 - \bar{G})\beta_7$$

Our null hypothesis can therefore be formulated as:

$$H_0 : \beta_1 - \bar{G}\beta_7 = 0 \text{ and } \beta_1 + (1 - \bar{G})\beta_7 = 0$$

This hypothesis is evaluated using a Wald test implemented by the `linearHypothesis` function.

```
# Calculate proportion of men in sample
G_bar <- mean(df$gender)

# Formulate null hypotheses
H0_women <- glue("D - {G_bar}*D:gender_c = 0")
H0_men <- glue("D + {1 - G_bar}*D:gender_c = 0")

linearHypothesis(lin_model,
                  hypothesis.matrix = c(H0_women,H0_men),
                  test = "Chisq")
```

Linear hypothesis test:

```
D - 0.49680170575693 D:gender_c = 0
D + 0.50319829424307 D:gender_c = 0
```

Model 1: restricted model

Model 2:  $Y \sim D + \text{age} + \text{gender} + \text{partyideology} + \text{pre\_therm\_voter\_outparty\_rescaled}$

```
Res.Df Df  Chisq Pr(>Chisq)
1      461
2      459  2 31.728 1.289e-07 ***
```

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

The results above indicate that we can be highly confident ( $p < 1 \times 10^{-6}$ ) that the treatment has a nonzero effect among either women or men.

## Part F

Here, we examine the moderating effect of baseline warmth towards outpartisans using (1) Lin's regression and (2) kernel regression.

### Lin's regression

In Part D, we fit a Lin's regression model that has the following form:

$$Y_i = \beta_0 + \beta_1 D_i + \dots + \beta_9 (D_i \times (W_i - \bar{W}))$$

where  $w_i$  is the baseline (pre-treatment) measurement of warmth towards outpartisans and  $\bar{w}$  is the average value of this covariate across the sample. The CATE given the value of this covariate is given by:

$$\text{CATE}(W = w) = \beta_1 + (w - \bar{W}) \beta_9$$

We can visualize this by plotting the CATE across the range of values observed in our data.

```
# Get coefficient estimates
beta1 <- lin_model$coefficients["D"]
beta9 <- lin_model$coefficients["D:pre_therm_voter_outparty_rescaled_c"]
W_bar <- mean(df$pre_therm_voter_outparty_rescaled)

# Get range of values
W_min <- min(df$pre_therm_voter_outparty_rescaled)
W_max <- max(df$pre_therm_voter_outparty_rescaled)
W_vals <- seq(W_min, W_max, length.out=200)

# Calculate CATE over range of values
CATE_W <- beta1 + beta9*(W_vals - W_bar)

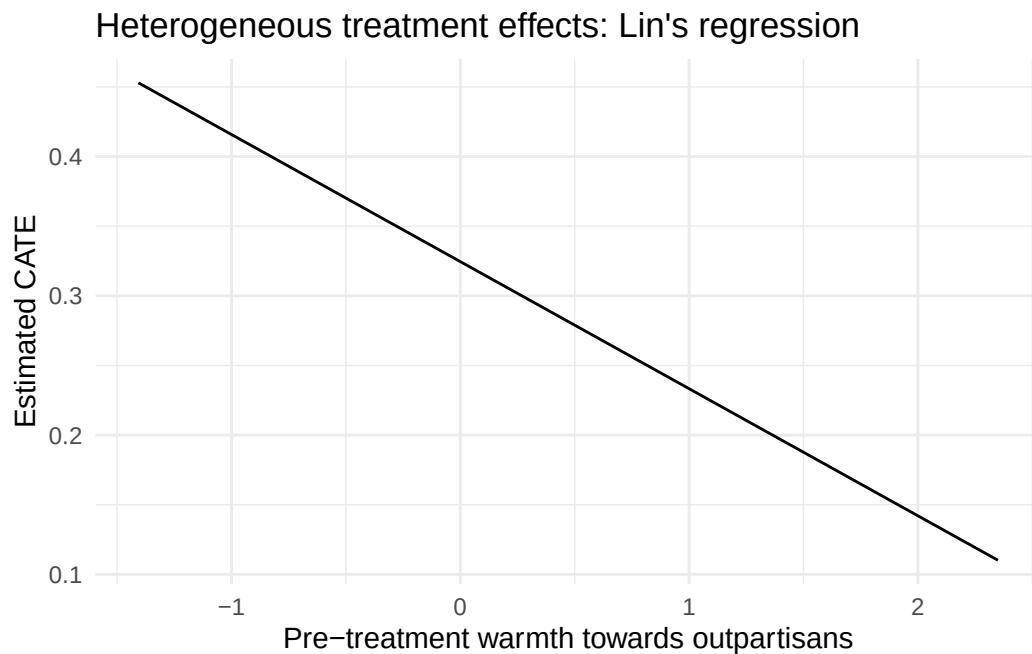
# Create plot
p <- ggplot(data.frame(w=W_vals, CATE=CATE_W), aes(x = w, y = CATE)) +
  geom_line() +
  labs(
    x = "Pre-treatment warmth towards outpartisans",
```

```

    y = "Estimated CATE",
    title = "Heterogeneous treatment effects: Lin's regression"
  ) +
  theme_minimal()

# Display to console
p

```



### Kernel regression

We can also visualize the moderating effect of pre-treatment warmth towards outpartisans using nonparametric techniques such as kernel regression.

```

## Fit kernel regression model with pre-treatment warmth as a moderator.

# Note that I'm manually setting the bandwidth because
# the default value appeared to be overfitting.
# Settled on a value of 0.3 based on the range of our moderator variable.

model_kernel <- interflex(
  Y = "Y",

```

```

D = "D",
X = "pre_therm_voter_outparty_rescaled",
data = df,
bw = 0.3,
estimator = "kernel",
vcov.type = "robust"
)

```

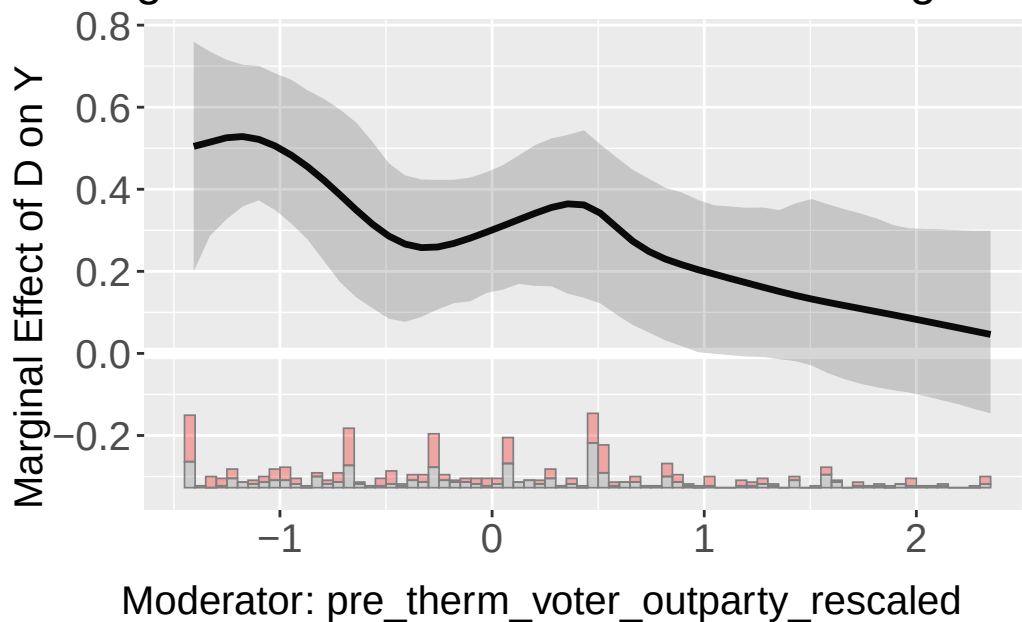
Baseline group not specified; choose treat = 0 as the baseline group.  
 Number of evaluation points:50  
 Parallel computing with 4 cores...

```

# Plot results
plot(model_kernel,
      main = "Heterogeneous treatment effects: Kernel regression")

```

## Heterogeneous treatment effects: Kernel regression



### Interpretation

The above results suggest that the treatment effect was reduced for those who had higher baseline measures of warmth towards outpartisans. One potential explanation for this is that

those who had higher baseline warmth towards outpartisans might already have regular face-to-face interactions with people whose political beliefs differ from theirs, which would make the experiment “just another day” for them. In contrast, those who had low baseline warmth might live in communities without much political diversity, which would make the experiment a more abnormal and impactful experience.